Complementary-Filtered Residual World Models for Multistep Prediction of Nonlinear Controlled Dynamics

Anonymous authors

Abstract

Accurate multistep prediction of nonlinear controlled dynamics depends on the forecast-origin state estimate and the transition model. We study the Complementary-Filtered Residual World Model (CFR-WM), which combines a fixed gain-0.5, mask-gated context update with a diagonal-Gaussian physics-residual transition. For correlated prediction and measurement errors, we derive the channel-wise mean-square-error optimal scalar gain. The confirmatory study trains 440 models across five nonlinear systems, three regimes, and eight seeds. Against the preselected physics-guided neural-network comparator, cell-balanced 25-step normalized root-mean-square error (NRMSE-25) improves by 8.97% (95% bootstrap interval [6.88%, 10.95%]; exact one-sided test against a 5% margin, $\boldsymbol{p = 0.00391}$). The aggregate hypothesis is supported, but clean-condition noninferiority fails. In secondary analyses, a classical gain-0.5 context filter with nominal rollout attains $\mathbf{0.111 \pm 0.003}$ NRMSE-25 versus $\mathbf{0.119 \pm 0.004}$ for CFR-WM; disabling the learned residual also gives the lowest mean among four inference scales. A synchronous two-step context outage increases CFR-WM error by 2.7%, versus 51.9% for the comparator. On held-out measured NanoDrone flights, CFR-WM has higher full-state NRMSE than a matched unfiltered residual model in all eight condition–horizon comparisons, but lower 50-step position error in all four; its nominal 90% coverage at 50 steps is 0.524–0.667. The results attribute the controlled-benchmark gain primarily to forecast-origin filtering and show the limitations of a uniform gain and diagonal uncertainty.

Keywords: nonlinear dynamics, state estimation, complementary filtering, physics-guided learning, multistep prediction, model predictive control

1 Introduction

Prediction models for nonlinear controlled systems support state estimation, receding-horizon optimization, fault diagnosis, and simulation-based control. Their accuracy depends on two coupled objects: the state supplied at the forecast origin and the transition used to propagate that state under future inputs. Learning-based model predictive control makes the same dependence explicit because model error enters each optimized prediction horizon [12]. Modern world models extend this idea with recurrent, latent, or transformer transitions and have demonstrated strong planning performance in continuous control and reinforcement learning [10, 11, 20, 29]. For physical systems, transition accuracy and state-estimation reliability are joint requirements when recent measurements are noisy or missing.

Recursive state estimation classically balances a dynamic prediction against new measurements. The Kalman filter computes this balance from a stochastic linear model and evolving error covariance [14], while complementary filters and nonlinear observers exploit structural separation between measurement channels and model dynamics [19, 30]. Learned filters such as KalmanNet relax parts of the assumed model, and data-assimilation research couples machine learning with dynamical estimation and uncertainty quantification [7, 23]. These methods motivate a direct question for multistep prediction: can an explicit,

fixed context correction improve a physics-residual transition when observations are incomplete or corrupted?

The answer depends on the relative reliability of the physical prediction and the measurement. Parameter drift, unmodeled friction, discretization, and actuator dynamics bias nominal propagation. Measurement noise, missing packets, and preprocessing errors bias or remove the observation. Last-value imputation supplies a numerical input but does not make it current; pure nominal propagation remains current but inherits model mismatch. Physics-guided learning combines mechanistic and neural components through constraints, features, hybrid solvers, and probabilistic state models [15, 25, 27, 28]. The forecast-origin correction nevertheless remains a distinct design choice from the learned transition.

We study this choice with the Complementary-Filtered Residual World Model (CFR-WM). Within an eight-step observed context, nominal dynamics predict the next state and a fixed gain of 0.5 combines that prediction with each available measurement. Missing channels receive zero measurement gain. The final corrected state initializes a probabilistic nominal-physics-plus-bounded-residual transition trained by multistep Gaussian likelihood and rolled out freely under known future controls. The update is a fixed-gain complementary filter that exposes the allocation between nominal prediction and measurement at the forecast origin.

The evaluation has two layers. First, five controlled nonlinear systems: a damped oscillator, Duffing oscillator, controlled Van der Pol oscillator, torque-driven pendulum, and cart–pole. They provide known equations, controlled parameter shift, and paired observation corruption. Eleven transition variants are compared across clean in-distribution, corrupted in-distribution, and corrupted parameter-shift regimes. The primary analysis averages the 15 system-regime cells within each of eight independent data-training seeds and tests a practical 5% improvement margin against a preselected physics-guided neural-network comparator. Fixed gains 0, 0.25, 0.5, 0.75, and 1 isolate the context correction, while a one-step-trained variant isolates the training horizon.

Second, an external experiment uses measured flight states from the NanoDrone benchmark [3]. Three flights from a held-out trajectory family are evaluated with a reduced quadrotor prior, three parameter-matched learned models, a nominal model, four context conditions, and 25- and 50-step horizons. Noise and missingness are varied separately and together. This experiment tests whether the controlled-benchmark finding persists when the physical prior is structurally incomplete and the targets are recorded flight states.

The main contributions are:

1. a mask-gated fixed-gain context correction coupled to a probabilistic physics-residual free rollout for nonlinear controlled dynamics;
2. an error identity for correlated nominal-prediction and measurement errors, its constrained MSE-optimal scalar gain, and fixed-gain sensitivity;
3. a paired confirmatory benchmark comprising 440 trained models, five systems, three regimes, and eight independent seed clusters; and
4. secondary classical-filter, residual-scale, structured-outage, projection, latency, and calibration audits, together with an external measured-flight evaluation.

The controlled benchmark supports the aggregate primary hypothesis against the preselected PGNN-style comparator but not the gated clean-regime noninferiority requirement. Classical component analysis and residual-scale sensitivity show that the gain-0.5 filter followed by nominal rollout performs better than the learned residual transition. The flight evaluation shows that lower long-horizon position error can coexist with higher full-state error and undercovered intervals. These component and uncertainty trade-offs matter when a predictor supplies an observer or predictive controller.

The remainder of the paper reviews state-estimation, physics-guided, and world- model approaches; defines the correction and free rollout; reports the controlled and flight experiments; and discusses implications for learned observers and receding-horizon prediction.

2 Related work

2.1 State estimation and complementary filtering

The Kalman filter established recursive prediction and measurement correction for linear stochastic systems, with gains determined by propagated covariance [14]. Complementary filters provide a lower-complexity alternative when measurements and model predictions have different reliability or frequency characteristics. Nonlinear complementary observers on SO(3) provide an influential example for attitude estimation [19]. Extended-state observers likewise estimate unmodeled interactions from motion measurements; recent IJDC work demonstrates this role for wheel–rail dynamics [30].

Learning has been introduced into this estimation loop in several ways. KalmanNet learns Kalman-style updates for partially known dynamics [23]; autodifferentiable ensemble Kalman filters expose the assimilation procedure to gradient-based learning [6]; hybrid estimators combine physical and data-driven components in energy and vehicle systems [4, 5]; and machine learning with data assimilation connects estimation to uncertainty quantification [7]. CFR-WM uses a fixed-gain mechanism: an observed channel receives a prespecified scalar gain, a missing channel receives zero gain, and the corrected state initializes a learned multistep predictor. A classical fixed-gain-filter-plus-nominal- rollout baseline separates the context filter from the neural residual.

2.2 Physics-guided dynamical learning and predictive control

Physics-guided learning can constrain outputs, embed equations, combine numerical solvers with neural corrections, or assimilate uncertain observations. Reviews emphasize that the appropriate integration depends on the dynamics, available measurements, and downstream objective [2, 8, 28]. Differentiable modeling exposes mechanistic components to gradient-based learning, while horizon-aware optimization addresses temporal error accumulation [17, 25, 26].

Physics-guided neural networks have been evaluated on nonlinear dynamics [24], and physics-guided Deep Markov Models combine state-space physics with probabilistic latent inference [18]. These approaches primarily shape the transition or latent dynamics. CFR-WM instead couples a fixed context correction to a bounded residual transition, allowing the forecast-origin and rollout contributions to be evaluated separately. This distinction also matters for learning-based model predictive control, where a learned prediction model is repeatedly queried across an optimization horizon and uncertainty or model mismatch can affect feasibility and performance [12]. Sparse system identification has likewise been coupled directly to nonlinear model predictive control in the low-data regime [13]. The present study focuses on the offline estimator–predictor mapping that supplies state forecasts to such controllers.

2.3 World models and predictive uncertainty

World models use recurrent stochastic state-space models, normalized latent representations, transformers, and other sequence architectures. DreamerV3 and TD-MPC2 demonstrate scalable learned dynamics for continuous control, while IRIS and STORM show transformer-based alternatives [10, 11, 20, 29]. For the explicit-state forecasting problem here, we implement compact, approximately parameter-matched recurrent, normalized, transformer, and physics-guided transition families under a common protocol. Broader surveys organize world models around representation, dynamics, planning, and agent learning [9].

Predictive uncertainty can represent observation noise, transition ambiguity, or model-form error, and likelihood alone does not establish calibrated intervals [1]. Dynamic- system studies assess coverage and interval width together, while Bayesian and conformal methods address model-form and distributional uncertainty [16, 21, 22]. We therefore report Gaussian negative log likelihood, empirical coverage, interval width, and calibration error in addition to multistep point accuracy.

3 Complementary-filtered residual world model

3.1 Forecasting problem

Let $\mathbf{x}_t \in \mathbb{R}^{d_x}$ be a physical state and $\mathbf{u}_t \in \mathbb{R}^{d_u}$ a recorded control. The observation available to the model is

$$\mathbf{y}_t = \mathbf{m}_t \odot (\mathbf{x}_t + \boldsymbol{\epsilon}_t), \tag{1}$$

where $\mathbf{m}_t \in \{0,1\}^{d_x}$ is a channel-wise availability mask and $\boldsymbol{\epsilon}_t$ is observation noise. Missing entries are filled causally by the latest available value, producing $\tilde{\mathbf{y}}_t$; the mask and elapsed time since observation $\boldsymbol{\delta}_t$ remain explicit inputs to distinguish imputed values from fresh measurements.

At forecast origin t, the model receives eight context states and their past controls,

$$\mathcal{C}_t = \{\tilde{\mathbf{y}}_{t-7:t}, \mathbf{m}_{t-7:t}, \boldsymbol{\delta}_{t-7:t}, \mathbf{u}_{t-7:t-1}\}, \tag{2}$$

together with known future controls $\mathbf{u}_{t:t+H-1}$. It estimates a diagonal predictive distribution for $\mathbf{x}_{t+1:t+H}$ under free rollout, with training horizon $H = 25$, using predicted states at subsequent rollout steps. The flight evaluation additionally extends rollout to $H = 50$. All learned operations use affine statistics estimated from the training split; the nominal equations are evaluated in physical coordinates and then returned to normalized coordinates.

3.2 Mask-gated complementary context filter

Let F_0 denote a nominal dynamics step: fourth-order Runge–Kutta integration for the controlled benchmark, and the reduced flight model in Section 6. Starting from the first imputed context vector, the filter alternates nominal prediction and measurement update. For context index $k > t-7$,

$$\bar{\mathbf{x}}_k = F_0(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1}), \tag{3}$$

$$\mathbf{K}_k = \alpha \mathbf{m}_k, \qquad \alpha = 0.5, \tag{4}$$

$$\hat{\mathbf{x}}_k = \Pi[\bar{\mathbf{x}}_k + \mathbf{K}_k \odot (\tilde{\mathbf{y}}_k - \bar{\mathbf{x}}_k)]\,. \tag{5}$$

Thus, before projection, an observed channel receives equal weight from the nominal prediction and the current measurement, while a missing channel has exactly zero measurement correction. The projection Π is the identity except for the sine–cosine pendulum representation, which is returned to the unit circle after each update. This joint projection couples the pendulum channels and can change a channel whose measurement gain is zero. The zero-gain statement therefore applies to the direct correction before projection, not necessarily to the projected state. The gain is fixed before training, independently of validation and test outcomes.

In parallel, a GRU encodes $[\tilde{\mathbf{y}}_k, \mathbf{m}_k, \log(1+\boldsymbol{\delta}_k), \mathbf{u}_{k-1}]$. Its last hidden state supplies trajectory context to the forecast transition; the measurement gain remains fixed. The implementation retains the same unused gain-head parameterization in every controlled-benchmark fixed-gain sensitivity model, so comparisons at gains 0, 0.25, 0.5, 0.75, and 1 do not change nominal capacity. This construction draws on filtering and data-assimilation approaches [7, 23] and uses the mask-gated context update to initialize a physics-residual probabilistic world model.

3.3 Probabilistic physics-residual rollout

The last filtered context state initializes a free rollout. At step h,

$$\boldsymbol{\mu}^p_{t+h} = F_0(\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}), \tag{6}$$

$$[\mathbf{r}_{t+h}, \ell_{t+h}] = R_\theta([\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}, \mathbf{b}_{t+h-1}]), \tag{7}$$

$$\boldsymbol{\mu}_{t+h} = \Pi\left[\boldsymbol{\mu}^p_{t+h} + 0.25\tanh(\mathbf{r}_{t+h})\right], \tag{8}$$

$$\boldsymbol{\sigma}^2_{t+h} = \max\left\{\exp(\mathrm{clip}(\ell_{t+h}, -8, 4))\,, 10^{-6}\right\}. \tag{9}$$

The fixed coefficient 0.25 bounds every per-step residual to one quarter of a training standard deviation in normalized coordinates and defines the residual scale convention used by the architecture. A fixed-weight sensitivity analysis varies this multiplier at inference to quantify its effect. The predicted mean becomes the next state, and a GRUCell updates $\mathbf{b}$ from that mean and the known control.

The training loss is the mean diagonal Gaussian negative log likelihood over all H steps and

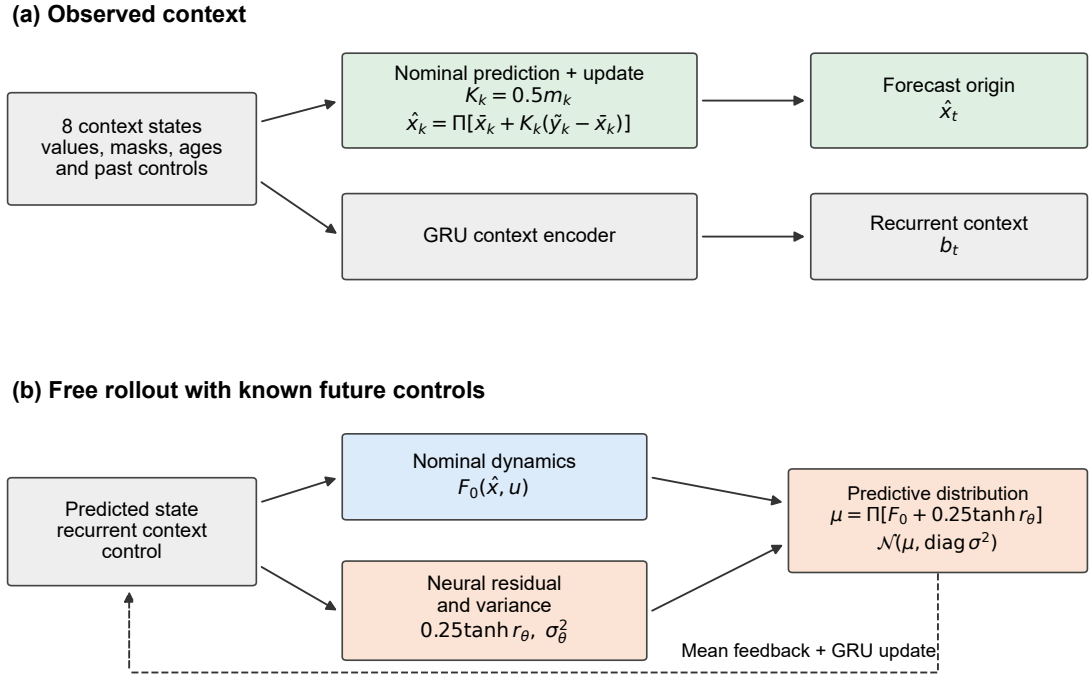


Fig. 1 Causal data flow of CFR-WM. A fixed complementary update is applied only inside the observed context. Missing channels receive zero direct measurement correction before any state-space projection. The filtered forecast origin then initializes a probabilistic nominal-physics plus neural-residual transition, which is rolled out freely for 25 steps using known future controls and no future observations.

state channels,

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{2Hd_x} \sum_{h=1}^{H} \sum_{q=1}^{d_x} \left[\log(2\pi\sigma_{hq}^2) + \frac{(x_{hq} - \mu_{hq})^2}{\sigma_{hq}^2} \right]. \tag{10}$$

The one-step ablation applies the same loss only at $h = 1$ but is still evaluated by a 25-step free rollout.

When a control objective supplies nonnegative state weights w_q, the same architecture admits the component-weighted loss

$$\mathcal{L}_w = \frac{1}{2H \sum_q w_q} \sum_{h=1}^{H} \sum_{q=1}^{d_x} w_q \times \left[\log(2\pi\sigma_{hq}^2) + \frac{(x_{hq} - \mu_{hq})^2}{\sigma_{hq}^2} \right]. \tag{11}$$

The experiments use $w_q = 1$ for all channels; Equation (11) states how a prespecified downstream cost could change that objective.

3.4 Causal execution and computational cost

For each window, execution follows four deterministic stages. First, the GRU encodes the imputed context together with masks, elapsed times, and past controls. Second, Equation (5) advances the nominal state through the same context and applies the mask-gated correction after each available observation. Third, the final hidden state and filtered state jointly initialize the residual recurrence. Fourth, Equation (9) is iterated for every known future control; predicted means, rather than samples or targets, feed the next step. Training backpropagates the multi-step NLL through this entire causal graph. Evaluation uses the identical graph with gradients disabled, so teacher forcing is absent from both validation and test rollouts.

Let T_c denote context length, H forecast horizon, d_b recurrent width, and C_{F_0} the cost of one nominal step. Dense recurrent operations cost $O((T_c+H)d_b^2)$ per trajectory, while nominal propagation costs $O((T_c + H)C_{F_0})$. Diagonal mean and variance storage is $O(Hd_x)$; recurrent working memory at inference is $O(d_b + d_x)$. Thus inference grows linearly in context and horizon. The fixed filter adds nominal context steps; measured latency captures their cost in the implementation.

3.5 Forecast-origin and rollout error

For an observed scalar channel on which Π is the identity, write the nominal pre-update error as $e_p = \bar{x} - x$ and measurement error as $\epsilon = y - x$. Before any non-identity projection, Equation (5) gives the exact post-update error

$$e_f = (1 - \alpha)e_p + \alpha\epsilon. \tag{12}$$

Let $v_p = \mathbb{E}[e_p^2]$, $v_\epsilon = \mathbb{E}[\epsilon^2]$, and $c = \mathbb{E}[e_p\epsilon]$ for zero-mean errors. Then

$$\mathbb{E}[e_f^2] = (1 - \alpha)^2 v_p + \alpha^2 v_\epsilon + 2\alpha(1 - \alpha)c. \tag{13}$$

Provided $v_p + v_\epsilon - 2c > 0$, the scalar gain minimizing this conditional MSE over $[0, 1]$ is

$$\alpha^* = \mathrm{clip}_{[0,1]}\left(\frac{v_p - c}{v_p + v_\epsilon - 2c}\right). \tag{14}$$

The fixed value 0.5 is optimal only when $v_p = v_\epsilon$ and $c = 0$. In an adaptive implementation, v_p, v_ϵ, and c can be estimated per channel from training or validation innovations and held fixed for test data, or predicted causally from sensor metadata. A missing channel has zero gain and therefore receives no direct measurement-error term. For the pendulum's joint sine–cosine projection, Equation (13) applies before projection; the projected error also includes the coupled projection contribution. These dependencies motivate the fixed-gain sensitivity analysis.

If the rollout transition is locally L-Lipschitz and incurs local error η_h, the standard recursion yields

$$\|\hat{\mathbf{x}}_{t+H} - \mathbf{x}_{t+H}\| \le L^H\|\hat{\mathbf{x}}_t - \mathbf{x}_t\| + \sum_{h=0}^{H-1} L^{H-1-h}\eta_h. \tag{15}$$

The context filter acts on the initial-error term, while the residual model must control subsequent local errors. The bound makes these two contributions to long-horizon forecasting explicit.

For receding-horizon control, Equations (5) and (9) define an estimator–predictor interface: the context update supplies the current state, and the residual transition maps a candidate future control sequence to a state distribution. Prediction error and channel-level sensitivity quantify the contribution of the fixed-gain interface in the following experiments.

3.6 Use of generative AI tools

Generative AI tools assisted literature screening, manuscript drafting, and software and analysis-script development. The authors verified the sources, equations, code, results, and scientific claims, approved the final manuscript, and take responsibility for its content.

4 Controlled benchmark design

4.1 Controlled systems, corruption, and shift

We evaluate five action-conditioned nonlinear systems. The damped oscillator has state $(q, \dot{q})$ and dynamics $m\ddot{q} + c\dot{q} + kq = u$; the Duffing oscillator follows $\ddot{q} + c\dot{q} + \alpha q + \beta q^3 = u$; and the controlled Van der Pol system follows $\ddot{q} = \mu(1-q^2)\dot{q}-q+g_u u$. The pendulum is represented by $(\sin\theta, \cos\theta, \dot{\theta})$, and the cart–pole by $(x, \dot{x}, \theta, \dot{\theta})$. Reference trajectories use fourth-order Runge–Kutta integration with four internal substeps. Initial conditions, bounded filtered stochastic controls, and physical parameters are generated from recorded seeds. Shifted parameter intervals are prespecified and disjoint from training intervals; unit tests verify that separation.

Each seed–system dataset contains 256 training, 64 validation, and 96 ID test trajectories, plus 96 independently generated OOD test trajectories. A trajectory has 72 states. Windows contain eight context states and 25 forecast transitions; training and validation strides are eight and 16, respectively, whereas evaluation uses one window per trajectory. Split identifiers encode system, parameter regime, split, trajectory index, and generation seed. No trajectory crosses splits.

The three evaluation regimes are (i) clean ID context, (ii) ID context with feature-wise Gaussian noise at 8% of the mean training-state standard deviation and a nominal 25% independent missing-at-random probability, and (iii) the same corruption with OOD physical parameters. Training context uses 3% noise and a nominal 10% missing probability. The first time point of every generated trajectory is forced observed in every channel so causal last-value imputation is defined; masks at all subsequent trajectory time points are sampled independently. A later sliding window can therefore begin at an unobserved, causally imputed context value. Masks and channel-wise elapsed times are retained. ID-clean and ID-corrupt evaluations reuse the same underlying 96 trajectories, permitting repeated-measurement resampling. OOD identifiers are required to be disjoint from ID identifiers.

4.2 Comparators and sensitivities

The five broad comparators cover recurrent, normalized, transformer, and physics-guided transition families. **RSSM-style** is a compact recurrent stochastic-style transition. **SimNorm-style** applies grouped simplex normalization to its recurrent representation, adapting a representation principle used by scalable continuous-control world models [11]. **Transformer-style** uses a two-layer causal context transformer followed by recurrent free rollout [20]. **PGNN-style** adds a deterministic neural correction to a nominal physics step [24]. **PG-DMM-style** is a probabilistic physics-residual transition inspired by physics-guided Deep Markov modeling [18]. These are protocol-matched transition implementations designed for the same explicit- state forecasting inputs and outputs.

The proposed **CFR-WM** uses gain 0.5. Four fixed-gain sensitivities use $\alpha \in \{0, 0.25, 0.75, 1\}$, and **CFR-one-step** changes only the training horizon. All six CFR variants have hidden width 72 and identical parameterization; even the inactive fixed-gain head is retained. Comparator widths are 80 for RSSM-style and SimNorm-style, 48 for Transformer-style, and 84 for PGNN-style and PG-DMM-style.

Architecture, gain, tuning budget, endpoints, and decision thresholds were fixed using development data before the eight confirmatory seeds were generated. Development trajectories and model outputs are excluded from the confirmatory analysis. This separation preserves a direct comparison between the gain-0.5 model, four fixed-gain sensitivities, and the one-step-trained variant.

4.3 Optimization and endpoint definitions

All models use AdamW, batch size 128, gradient clipping at norm 5, at most 60 epochs, and early stopping after ten epochs without a validation-NLL improvement of 10^{-5}. The pre-confirmation tuning archive contains the same six-trial validation budget for every baseline–system pair. Each baseline uses its own selected winner. CFR-WM and all CFR sensitivities inherit the selected PGNN-style winner for the same system. All hyperparameters were fixed before confirmation.

The primary endpoint is NRMSE through horizon 25. Per-feature RMSE is divided by that feature's standard deviation estimated from the training split only, then averaged over features. Because stored targets and predictions are already in those normalized coordinates, the evaluation divisor is one. This avoids estimating a denominator from the test trajectory and remains defined at horizon one. Secondary accuracy endpoints are cumulative NRMSE at horizons 1, 5, and 10 and MAE through horizon 25. Probabilistic endpoints are diagonal Gaussian NLL, empirical 90% coverage, mean 90% interval width, and mean absolute central-interval calibration error over nominal levels 0.1–0.9. A trajectory is catastrophic when any feature-step absolute error exceeds three training-set standard deviations.

4.4 Confirmatory inference

The confirmation grid contains 11 methods, eight untouched seeds $(41, 53, 67, 79, 83, 97, 109, 113)$, five systems, and three regimes: 440 training runs, 1,320 aggregate cells, and 126,720 paired trajectory rows. One deterministic dataset fingerprint is shared by all methods within each seed–system group, and all 40 such fingerprints must be distinct. The eight seed-level training/data realizations are the top-level independent clusters; trajectories are nested observations. Inference conditions on the fixed set of systems and regimes.

For a seed s, let B_s and C_s be the PGNN-style and CFR-WM NRMSE-25 means after equal weighting of all 15 system–regime cells. The primary estimand is

$$\Delta_{\mathrm{rel}} = \frac{1}{8}\sum_{s=1}^{8} 100\frac{B_s - C_s}{B_s}. \tag{16}$$

A 5,000-replicate bootstrap resamples seeds globally with replacement. Within each selected seed and system it resamples paired trajectory identifiers; ID-clean and ID-corrupt reuse the same sampled identifier vector, while the disjoint OOD set is resampled independently. Every system–regime cell is retained, and B_s, C_s, and their ratio are recomputed in each replicate. The interval is the two-sided percentile 95% interval.

H1 states that CFR-WM improves on the preselected PGNN-style comparator by at least 5%. Support requires finite metrics, a lower relative-improvement interval endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the seed effects after subtracting the 5% margin. All 2^8 sign configurations are enumerated under a prespecified symmetry assumption. Confirmation success is determined by H1 alone.

H2 is tested only if H1 succeeds. It requires at least 5% improvement when pooling ID-corrupt and OOD-corrupt fixed cells, and clean-regime noninferiority with signed relative-improvement margin -5%. Both components must meet their respective bootstrap bound and one-sided margin test, with all metrics finite. The five broad CFR-versus-comparator contrasts form one Holm family; the five CFR sensitivity/ablation contrasts form another. With eight clusters, the minimum two-sided exact raw and five-test Holm-adjusted values are $2/2^8 = 0.0078125$ and 0.0390625. The 15 system–regime contrasts report paired effects and unadjusted bootstrap intervals as descriptive estimates.

4.5 Secondary component and stress analyses

Six secondary analyses characterize the fitted models. First, two nonlearned rollouts use the nominal dynamics from either the final imputed context state or the gain-0.5 filtered context state. Second, CFR-WM is evaluated with residual multipliers 0, 0.125, 0.25, and 0.5 while holding the trained weights fixed; the stored 0.25 predictions are reproduced exactly. Third, a synchronous two-step all-channel outage is inserted at context indices 3–4, 5–6, or 7–8 of otherwise clean ID windows and the three start positions are averaged. Fourth, the pendulum audit measures how often unit-circle projection changes a missing sine or cosine channel after its zero direct correction. Fifth, stored RTX 5070 batch timings are summarized. Each timing cell places 96 trajectories in one batch and includes CPU-to-GPU transfer, context encoding, and 25-step rollout, but excludes dataloader iteration and metric computation. Sixth, external-flight NLL and marginal interval metrics are recomputed from all 15 final learned checkpoints. These analyses are descriptive and use the original seed and cell weighting.

4.6 Reproducibility controls

The reproducibility contract fixes code hashes, configurations, data identities, trajectory pairing, and finite-metric checks. Analysis proceeds only after the complete grid and stored provenance pass these checks.

Execution uses Python 3.11.15 and PyTorch 2.11.0+cu130 on an NVIDIA RTX 5070 Laptop GPU with 8,151 MiB reported memory and an AMD Ryzen 9 9955HX processor.

5 Controlled benchmark results

5.1 Aggregate accuracy

The analysis includes the complete grid of 440 trained models and 126,720 trajectory-level rows. All prespecified metrics were finite (yes), and the 40 seed–system groups produced 40 distinct data fingerprints while sharing one fingerprint across methods within each group.

Table 1 reports the seed-balanced mean and standard deviation for horizons 1, 10, and 25. Across all 11 methods, CFR-WM had the lowest aggregate NRMSE-25 (0.0998). CFR-WM's mean was 0.0998, compared with 0.1096 for the preselected PGNN-style comparator.

For the primary estimand, PGNN-style and CFR-WM NRMSE-25 are first averaged with equal weight over the 15 prespecified system–regime cells; their relative improvement is then

Table 1 Confirmatory accuracy over the fixed five-system, three-regime benchmark. Entries are means ± SD across eight seed-level, cell-balanced aggregates. Bold marks the lowest aggregate NRMSE-25. Parameter counts are means over the five system-specific input dimensions; all methods received the same six-trial validation search budget.

Method	Mean parameters	NRMSE-1	NRMSE-10	NRMSE-25
CFR-WM	57694	0.039 ± 0.001	0.057 ± 0.002	**0.100 ± 0.004**
CFR-gain-0.25	57694	0.037 ± 0.001	0.059 ± 0.002	0.103 ± 0.004
CFR-gain-0.75	57694	0.047 ± 0.001	0.065 ± 0.002	0.107 ± 0.005
PGNN-style	61275	0.060 ± 0.001	0.070 ± 0.002	0.110 ± 0.003
CFR-gain-1	57694	0.059 ± 0.001	0.073 ± 0.002	0.114 ± 0.004
PG-DMM-style	61493	0.063 ± 0.001	0.075 ± 0.002	0.114 ± 0.003
RSSM-style	62485	0.073 ± 0.002	0.090 ± 0.005	0.123 ± 0.006
Transformer-style	59708	0.074 ± 0.002	0.094 ± 0.004	0.132 ± 0.009
SimNorm-style	62485	0.074 ± 0.002	0.098 ± 0.006	0.138 ± 0.013
CFR-gain-0	57694	0.081 ± 0.003	0.100 ± 0.003	0.144 ± 0.005
CFR-one-step	57694	0.040 ± 0.001	0.136 ± 0.050	0.278 ± 0.104

computed. The confirmatory point estimate is the arithmetic mean of those eight seed-level relative improvements rather than the ratio of the two pooled means in Table 1.

5.2 Primary and gated robustness hypotheses

Against PGNN-style, the seed-mean relative NRMSE-25 improvement was 8.97% (95% paired fixed-cell bootstrap CI [6.88%, 10.95%]). In absolute units, CFR-WM minus PGNN-style was -0.0098 (95% CI [-0.0121, -0.0075]). The exact one-sided sign-flip test against the same 5% practical margin gave $p = 0.00391$. Under the prespecified conjunction of a 5% interval lower bound, $p < 0.05$, and finite metrics, H1 was supported.

For the H2 components, the combined corrupt-ID and corrupt-OOD regimes had a relative effect of 12.05% (95% CI [10.51%, 13.49%], one-sided 5%-margin $p = 0.00391$). ID-clean had an effect of -13.08% (95% CI [-20.60%, -5.67%], one-sided signed -5%-margin $p = 0.95703$). Because H2 is hierarchically gated by H1 and requires both component rules, the robustness hypothesis was not supported.

Table 2 reports the eight independent cluster effects. Every seed exceeded the 5% practical threshold.

Table 2 Seed-level primary effects after equal weighting of the 15 benchmark cells. Positive values favor CFR-WM.

Seed	CFR-WM improvement over PGNN-style, %
41	+5.29
53	+12.76
67	+12.01
79	+10.49
83	+9.69
97	+6.66
109	+9.32
113	+5.56

5.3 Comparator and sensitivity evidence

Table 3 reports every prespecified broad comparison. Positive relative effects favor CFR-WM. The confidence intervals retain all fixed cells, whereas the exact two-sided seed sign-flip values are Holm-adjusted as one five-test family. These secondary comparisons contextualize the primary PGNN-style contrast.

The PGNN-style row repeats the primary point estimate but uses the comparator-family bootstrap seed (20260808), whereas H1 uses its prespecified primary seed (20260805). The estimand and 5,000-replicate resampling scheme are otherwise identical; the separately sampled interval endpoints differ by at most 0.01 percentage point after rounding, and only the primary interval enters the H1 decision.

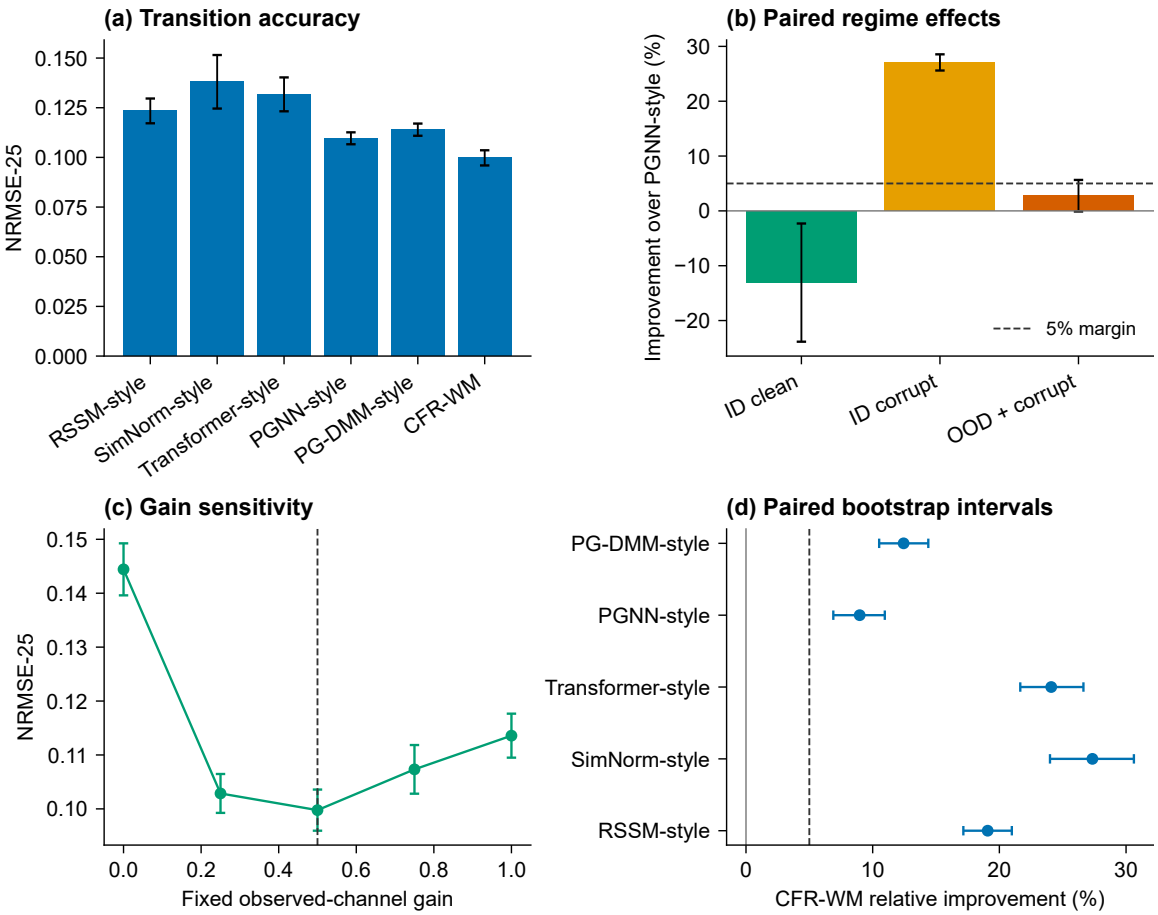


Fig. 2 Confirmatory summary. (A) Aggregate transition accuracy for the six main methods. (B) Paired CFR-WM effects against PGNN-style by regime; positive values favor CFR-WM. (C) Prespecified fixed-gain sensitivity. (D) Paired fixed-cell intervals against each broad comparator. Error bars in A–C are seed-level SD; D uses the nested 95% bootstrap interval.

Table 3 CFR-WM relative NRMSE-25 effects against broad comparators. Exact p values test a zero seed-level difference; Holm adjustment is confined to this prespecified family of five.

Reference	Relative effect [95% CI], %	Exact p	Holm p
RSSM-style	+19.08 [+17.16, +20.98]	0.0078	0.0391
SimNorm-style	+27.34 [+24.00, +30.62]	0.0078	0.0391
Transformer-style	+24.09 [+21.65, +26.64]	0.0078	0.0391
PGNN-style	+8.97 [+6.89, +10.96]	0.0078	0.0391
PG-DMM-style	+12.44 [+10.51, +14.39]	0.0078	0.0391

The lowest aggregate NRMSE-25 among gains 0, 0.25, 0.5, 0.75, and 1 occurred at gain 0.50. Relative to the one-step-trained variant, CFR-WM's effect was 59.38% (95% CI [48.88%, 68.88%], Holm $p = 0.03906$). Table 4 gives the complete second five-test family.

Table 5 reports secondary diagnostics of Gaussian forecast quality, marginal calibration, and catastrophic-rollout frequency.

5.4 Fixed-cell heterogeneity

Figure 3 retains all 15 system–regime effects. By the prespecified descriptive guard, CFR-WM was

Table 4 CFR-WM relative NRMSE-25 effects against gain and training-horizon sensitivities. Positive values favor the proposed gain-0.5 multi-step model.

Reference	Relative effect [95% CI], %	Exact p	Holm p
Gain 0	+30.89 [+28.95, +32.67]	0.0078	0.0391
Gain 0.25	+2.97 [+0.88, +5.27]	0.0312	0.0391
Gain 0.75	+6.97 [+4.77, +9.42]	0.0078	0.0391
Gain 1	+12.13 [+10.31, +13.92]	0.0078	0.0391
One-step training	+59.38 [+48.88, +68.88]	0.0078	0.0391

Table 5 Secondary aggregate diagnostics for the six broad methods. Rates and scores are seed-balanced over all fixed cells and are not separately tested.

Method	Parameters	Gaussian NLL	Calibration error	Catastrophic rate
RSSM-style	62485	-0.183	0.214	0.01%
SimNorm-style	62485	-0.246	0.206	0.01%
Transformer-style	59708	-0.279	0.202	0.01%
PGNN-style	61275	-0.385	0.248	0.00%
PG-DMM-style	61493	0.029	0.245	0.00%
CFR-WM	57694	0.013	0.228	0.00%

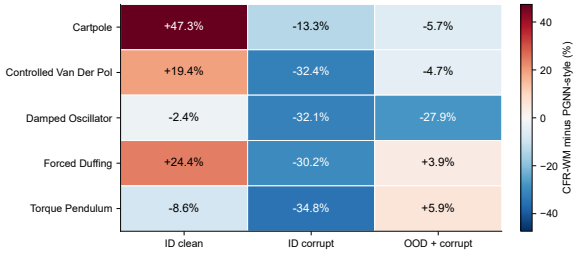


Fig. 3 Descriptive CFR-WM change relative to PGNN-style in each fixed benchmark cell. Negative values mean lower NRMSE-25 for CFR-WM.

more than 10% worse than PGNN-style in 3 cells. All three were ID-clean: cart–pole (+47.3% relative change), controlled Van der Pol (+19.4%), and Duffing (+24.4%). Conversely, 5 of 5 ID-corrupt cells favored CFR-WM, with descriptive changes from -13.3% to -34.8%. This pattern explains why an aggregate H1 success coexists with the clean-regime noninferiority result. These cell-level estimates are descriptive; their unadjusted bootstrap intervals are reported in the appendix table and machine-readable CSV.

5.5 Component analyses and stress tests

Table 6 separates the context filter from the learned residual. The gain-0.5 filter followed by nominal rollout reaches 0.111 NRMSE-25, below both CFR-WM (0.119) and PGNN-style (0.132). Evaluating the trained CFR-WM weights at their stored residual multiplier of 0.25 reproduces the stored cell means with maximum absolute difference 0.00e+00. Across the four tested inference multipliers, the lowest aggregate and regime-specific means occur at scale 0. The learned residual therefore increases the fixed-weight mean error in this analysis.

The synchronous two-step all-channel outage produces a 2.7% increase over clean-context CFR-WM, compared with 51.9% for PGNN-style (Table 7). For the pendulum, unit-circle projection changes 75.4% of the 5404 missing sine/cosine entries after the direct correction. The mean, 95th percentile, and maximum absolute normalized changes are 0.0328, 0.1374, and 0.3848, respectively. Thus the mask sets the direct measurement gain to zero, but projection can still alter a missing coordinate.

Table 6 Classical baselines and inference-time residual-scale analysis. Entries are seed-balanced mean NRMSE-25 across the indicated fixed cells. The scale analysis holds trained CFR-WM weights fixed and isolates the residual contribution.

Analysis and variant	All	Clean	ID corrupt	OOD corrupt
Nominal, final imputation	0.137	0.040	0.151	0.221
Complementary filter + nominal	0.111	0.042	0.092	0.201
PGNN-style	0.132	0.052	0.134	0.211
CFR-WM	0.119	0.055	0.097	0.204
CFR-WM, inference scale 0	0.111	0.042	0.092	0.201
CFR-WM, inference scale 0.125	0.113	0.046	0.093	0.202
CFR-WM, inference scale 0.25	0.119	0.055	0.097	0.204
CFR-WM, inference scale 0.5	0.135	0.079	0.113	0.213

Table 7 Clean-ID performance after a synchronous two-step all-channel context outage. Means ± SD and relative changes are across eight seed-level aggregates after averaging systems and the three outage positions.

Method	Outage NRMSE-25	Change from clean (%)
PGNN-style	0.079 ± 0.004	+51.9
CFR-WM	0.057 ± 0.003	+2.7

Table 8 Measured 25-step rollout time on the RTX 5070. Each cell uses one batch of 96 trajectories and includes transfer, context encoding, and rollout; SD is across seed-level cell-balanced means.

Method	ms/trajectory	SD
RSSM-style	0.110	0.006
SimNorm-WM	0.113	0.005
Transformer-style	0.109	0.007
PGNN-style	0.347	0.037
PG-DMM-style	0.344	0.036
CFR-WM	0.416	0.015

Table 8 reports the stored measured runtime. CFR-WM requires 0.416 ± 0.015 ms per trajectory, including its nominal context filtering. The measurement is a throughput statistic for one 96-trajectory GPU batch, not single-trajectory online latency.

6 External evaluation on measured drone flights

6.1 Data, held-out trajectories, and physical prior

We extend the controlled study to the NanoDrone benchmark, which records Crazyflie 2.1 Brushless flights using motion capture and onboard sensors [3]. The released records include the dataset authors' offline synchronization, 100 Hz resampling, motor–acceleration alignment, and zero-phase low-pass filtering. We retain these records and predict their recorded state estimates. Position, velocity, rotation-vector orientation, and body angular velocity form a 12-coordinate state; four motor angular speeds are the inputs.

Complete flights define the split. Runs 1–3 of Square, Random, and Chirp provide nine training records (41,700 samples); run 4 of each supplies three validation records (13,899 samples). Three Melon flights (19,497 samples) test a trajectory family absent from training. This validation hold-out is an addition to the authors' training/test partition. Windows remain within flights, and normalization uses training samples only.

The nominal model combines gravity with thrust along the body vertical axis:

$$a_t = c_T R(r_t) e_3 \sum_{j=1}^{4} \omega_{j,t}^2 - g e_3. \tag{17}$$

The kinematic updates are $v_{t+1} = v_t + \Delta t\, a_t$ and $p_{t+1} = p_t + \Delta t\, v_t + \frac{1}{2}\Delta t^2 a_t$. Here r_t is the rotation vector, $\omega_{j,t}$ is motor speed, $g = 9.81\,\mathrm{m\,s^{-2}}$, and $\Delta t = 0.01\,\mathrm{s}$. A nonnegative least-squares fit on training transitions gives $c_T = 8.31887655 \times 10^{-7}$ in $(\mathrm{m\,s^{-2}})/(\mathrm{rad\,s^{-1}})^2$. Orientation advances by quaternion composition under constant body angular velocity. The prior omits angular acceleration, drag, and motor lag, adding structural mismatch to the parameter mismatch studied in the simulations.

6.2 Matched comparisons and evaluation protocol

The flight implementation uses a width-64 recurrent encoder and residual transition. Three learned models share 47000 parameters: CFR-WM with gain 0.5, an unfiltered physics-residual model, and a data-only recurrent transition. The latter replaces the nominal physics step with a learned increment; the unfiltered model starts from the final imputed measurement. All three use the same encoder, bounded correction, and diagonal-Gaussian likelihood. The context GRU receives 40 features; a width-64 GRU cell advances the hidden state from 12 predicted coordinates and four inputs. An $80 \to 64 \to 64 \to 24$ SiLU head outputs 12 corrections and 12 log variances. Corrections use $0.25\tanh(\cdot)$ in normalized coordinates, and log variances are clipped to $[-8, 4]$. A nominal-only forecast supplies a fourth reference. These matched implementations follow the physics, hybrid, and recurrent families represented in the NanoDrone benchmark.

Each window contains eight context states. Last-value imputation and elapsed missingness times are computed within each flight before window extraction, so an imputed entry can retain an observation preceding the eight-state context. Training uses an equal mixture of recorded contexts and contexts with Gaussian noise of 0.05 training standard deviations per coordinate and 10% independent missingness. Controls and targets are unchanged. All learned models train on 25-step rollouts with stride 10, AdamW, batch size 256, at most 40 epochs, and eight-epoch early stopping. Each method selects between learning rates 0.001 and 0.003 by validation NLL using development seed 17, then trains with five final seeds (131, 149, 167, 181, 199).

Evaluation uses four paired context conditions: as recorded, Gaussian noise only (0.1 training standard deviations), missingness only (30%), and their combination. The same perturbation realization is shared across methods and training seeds. Known recorded motor inputs drive free forecasts through 25 and 50 steps (0.25 and 0.5 s), with test stride 50. For flight f, model m, and horizon H, the external score is

$$E_{m,f,H} = \left[\frac{1}{12 H N_f} \sum_{i=1}^{N_f} \sum_{h=1}^{H} \sum_{d=1}^{12} \left(\frac{\widehat{x}_{m,i,h,d} - x_{i,h,d}}{s_d} \right)^2 \right]^{1/2}, \tag{18}$$

where s_d is the training standard deviation. This score takes one root after pooling normalized squared errors; the controlled study instead averages feature-wise normalized RMSE. External summaries first give the three flights equal weight, then report the mean and sample SD across five training seeds. Position and velocity scores use root-mean-square Euclidean error in physical units. Paired relative improvements are averaged across seeds after the same flight weighting.

The external comparisons use no controlled-benchmark test data for training or model selection and are reported descriptively.

6.3 Flight forecasting results

Table 9 reports all eight condition–horizon comparisons. Each learned model has lower mean full-state NRMSE than the nominal forecast. The unfiltered residual model has lower mean full-state NRMSE than CFR-WM in all eight comparisons. With recorded contexts at 50 steps, their scores are 0.518 and 0.530, respectively. Combined noise and missingness narrows this difference to 0.534 versus 0.540. CFR-WM improves over the data-only recurrent model at 25 steps under added noise

and combined corruption, but its mean error is higher at 50 steps in all four conditions.

The physical-unit scores distinguish position and velocity behavior (Table 10; Figure 4). CFR-WM has the lowest 50-step position error among the three learned models in all four conditions. Under combined corruption, its position error is 0.094 m, compared with 0.125 m for the unfiltered model. The nominal prior has lower position error than the learned models on recorded and missing-only contexts, despite its higher full-state error. Velocity errors show a different ordering: CFR-WM is worse than the unfiltered model on recorded and missing-only contexts, and nearly tied under noise and combined corruption. Thus, the position benefit coexists with a full-state accuracy cost on these held-out flights.

The 50-step probabilistic diagnostics quantify forecast calibration (Table 11). CFR-WM has higher Gaussian NLL, lower nominal 90% coverage, narrower intervals, and higher calibration error than the unfiltered residual model in every context condition. Its coverage ranges from 0.524 to 0.667, well below 0.9. The diagonal likelihood therefore produces undercovered marginal intervals on these held-out flights; lower position error does not imply reliable joint-state uncertainty.

7 Discussion

The confirmatory analysis supports the prespecified practical-gain claim. Against the preselected PGNN-style comparator, the lower bootstrap endpoint clears the 5% practical margin and the exact seed-level test rejects that margin. The classical baseline refines the interpretation of this result. A gain-0.5 context filter followed by the nominal transition reaches 0.111 NRMSE-25, compared with 0.119 for CFR-WM and 0.132 for PGNN-style. The confirmatory H1 remains supported for its prespecified comparison. The component analysis attributes the aggregate advantage to forecast-origin filtering rather than to the learned residual transition.

The fixed-weight scale analysis strengthens that attribution. Removing the residual at inference gives the lowest mean in clean ID, corrupt ID, corrupt OOD, and the aggregate; increasing the multiplier from 0 to 0.5 monotonically increases aggregate NRMSE-25. With transition weights held fixed, the fitted correction increases the reported cell-balanced point error. The nominal prior and context correction capture the useful structure in this benchmark.

The regime decomposition explains when the filter helps relative to PGNN-style. The corrupted and shifted regimes improve by 12.05%, whereas the clean regime changes by -13.08%; consequently H2 does not support the gated robustness claim. The synchronous two-step outage produces only a 2.7% increase over clean-context CFR-WM, versus 51.9% for PGNN-style. This extends the missing-at-random evidence to a short, synchronized loss of all channels and indicates that nominal context propagation is valuable during brief sensor interruptions.

Equation (14) turns the gain trade-off into an estimator design rule. Equal trust is optimal only for equal prediction and measurement error variances with zero cross-covariance. Per-channel estimates of v_p, v_ϵ, and c from training or validation innovations would produce fixed channel gains without using test outcomes; a causal network could instead condition them on current sensor metadata. Either construction matches the observed heterogeneity better than one scalar gain. The pendulum audit also shows why gain semantics and state constraints must be separated: 75.4% of missing sine/cosine entries change under the subsequent unit-circle projection, even though their direct measurement correction is exactly zero.

The measured-flight evaluation exposes a state-component and uncertainty trade-off. The matched unfiltered residual model has lower mean full-state NRMSE in every condition–horizon comparison, whereas CFR-WM has lower 50-step position error in all four conditions. On recorded contexts, velocity error rises from 0.233 to 0.249 $\mathrm{m\,s^{-1}}$. At the same horizon, CFR-WM's nominal 90% marginal coverage is only 0.524–0.667 and is below the unfiltered model in all four conditions. Narrower intervals accompany higher NLL and calibration error, consistent with missing cross-state dependence and unrepresented model-form error in the diagonal Gaussian forecast.

These results connect estimator design to the control objective. Uniform NLL weights every normalized state component equally, while a controller can value position, velocity, or constraint-related coordinates differently. Equation (11) makes such weights explicit and requires them to

Table 9 External flight NRMSE (Equation 18). Entries are means ± SD across five training seeds after equal weighting of three held-out flights. The nominal forecast is deterministic. All four context conditions and both horizons are retained.

Context	Horizon	CFR-WM	Unfiltered	Recurrent	Nominal
As recorded	25	0.389 ± 0.012	0.376 ± 0.010	0.388 ± 0.008	0.488
As recorded	50	0.530 ± 0.016	0.518 ± 0.019	0.519 ± 0.006	0.790
Noise	25	0.393 ± 0.012	0.387 ± 0.010	0.398 ± 0.008	0.498
Noise	50	0.532 ± 0.017	0.523 ± 0.018	0.527 ± 0.006	0.797
Missing	25	0.397 ± 0.011	0.386 ± 0.008	0.395 ± 0.005	0.495
Missing	50	0.537 ± 0.014	0.527 ± 0.020	0.526 ± 0.007	0.792
Noise + missing	25	0.401 ± 0.011	0.397 ± 0.008	0.405 ± 0.005	0.506
Noise + missing	50	0.540 ± 0.014	0.534 ± 0.019	0.533 ± 0.006	0.799

Table 10 External 50-step position and velocity error in physical units, using the same flight and seed weighting as Table 9.

	Position RMSE (m)		Velocity RMSE (m/s)	
Context	CFR-WM	Unfiltered	CFR-WM	Unfiltered
As recorded	0.078 ± 0.003	0.083 ± 0.010	0.249 ± 0.015	0.233 ± 0.009
Noise	0.091 ± 0.003	0.115 ± 0.007	0.257 ± 0.014	0.258 ± 0.007
Missing	0.081 ± 0.005	0.095 ± 0.022	0.259 ± 0.014	0.243 ± 0.009
Noise + missing	0.094 ± 0.004	0.125 ± 0.017	0.267 ± 0.014	0.268 ± 0.009

be fixed from the downstream specification rather than selected from test performance. Correlated covariance or calibrated prediction sets would also be needed when constraints depend jointly on several state coordinates.

The measured runtime places CFR-WM at 0.416 ms per trajectory for an eight-step context and 25-step rollout on the stated GPU and represents throughput for 96 simultaneous trajectories. Online control requires separate timing and closed-loop tests for a specified controller, plant, sampling deadline, and disturbance model [12].

8 Limitations

The controlled evidence concerns five equation families with known inputs and eight independent data-training seed clusters. The secondary component analyses use the same trained checkpoints and are descriptive. The structured missingness test is a synchronous two-step outage inside an eight-step context. Longer bursts, informative dropout, sensor- specific failure, and delayed measurements are not represented.

The measured-flight evidence contains three held-out trajectories from one platform and collection session. Targets are the dataset authors' offline-preprocessed state estimates, and forecasts use recorded future motor inputs. The resulting scope is offline prediction on one platform and collection session; raw-sensor estimation, cross-platform transfer, and closed-loop control require separate evaluation.

The flight prior omits angular acceleration, drag, and motor lag. A uniform fixed gain cannot track channel- or time-dependent reliability. The diagonal-Gaussian output omits cross-state covariance and multimodality, and its 50-step marginal intervals under-cover on the held-out flights. Unit-circle projection couples pendulum coordinates after the mask-gated correction. No stability certificate follows from the bounded residual or the reported open-loop prediction errors.

Table 11 External 50-step probabilistic diagnostics. Coverage and width refer to marginal nominal 90% Gaussian intervals; calibration error averages absolute central-interval coverage error over nominal levels 0.1–0.9. Entries are means across five seeds after equal weighting of the three held-out flights.

Condition	Method	NLL	Cov.	Width	Cal. error
Recorded	CFR-WM	2.264	0.608	0.453	0.178
Recorded	Unfiltered residual	0.794	0.716	0.526	0.099
Noise	CFR-WM	2.994	0.524	0.454	0.242
Noise	Unfiltered residual	2.361	0.565	0.528	0.217
Missing	CFR-WM	1.829	0.667	0.534	0.123
Missing	Unfiltered residual	0.735	0.779	0.685	0.053
Combined	CFR-WM	2.200	0.613	0.535	0.184
Combined	Unfiltered residual	1.130	0.703	0.688	0.124

9 Conclusion

CFR-WM combines a gain-0.5 context filter with nominal-physics and learned- residual rollout. Against the prespecified PGNN-style comparator, it improves cell-balanced NRMSE-25 by 8.97% (95% CI [6.88%, 10.95%]), so H1 is supported. H2 is not supported because clean-condition noninferiority fails.

The classical baseline identifies the source of the benchmark result. The gain-0.5 filter followed by nominal rollout records 0.111 NRMSE-25, below CFR-WM's 0.119, and residual scale zero is best in every reported regime. The benchmark gain therefore arises from forecast-origin filtering; the tested residual adds no point-accuracy benefit. A short synchronous all-channel outage increases CFR-WM error by 2.7%, versus 51.9% for PGNN-style.

On held-out flights, CFR-WM lowers 50-step position error in all four context conditions but raises full-state NRMSE in all eight condition–horizon comparisons and yields undercovered 50-step intervals. Together, these findings motivate channel-specific gains derived from prediction–measurement error statistics, state weights tied to the control objective, and covariance models that represent cross-state uncertainty.

Appendix A Benchmark specification

Table A1 records the simulator time step and nominal parameterization; Table A2 gives every sampled training and OOD interval. All OOD intervals are non-overlapping on every named shifted parameter. These ranges define the fixed benchmark.

The action generator is a bounded first-order filtered process. At step t, a Gaussian innovation with scale 0.55 times the system action limit is combined with the previous action using coefficients 0.18 and 0.82, then clipped. Limits are 1.5, 1.8, 1.2, 2.2, and 8.0 for the systems in Table A1. This produces persistent excitation independently of the simulated state and defines the control-input distribution.

Train/validation/test separation occurs before window construction. A root NumPy `SeedSequence` spawns distinct split generators, which in turn spawn one generator per trajectory. The OOD test has a separate root seed. Observation corruption uses another recorded seed after state generation. Consequently, changing an observation-noise level does not change the underlying clean ID trajectory, and all methods see exactly the same corruption draw within a condition.

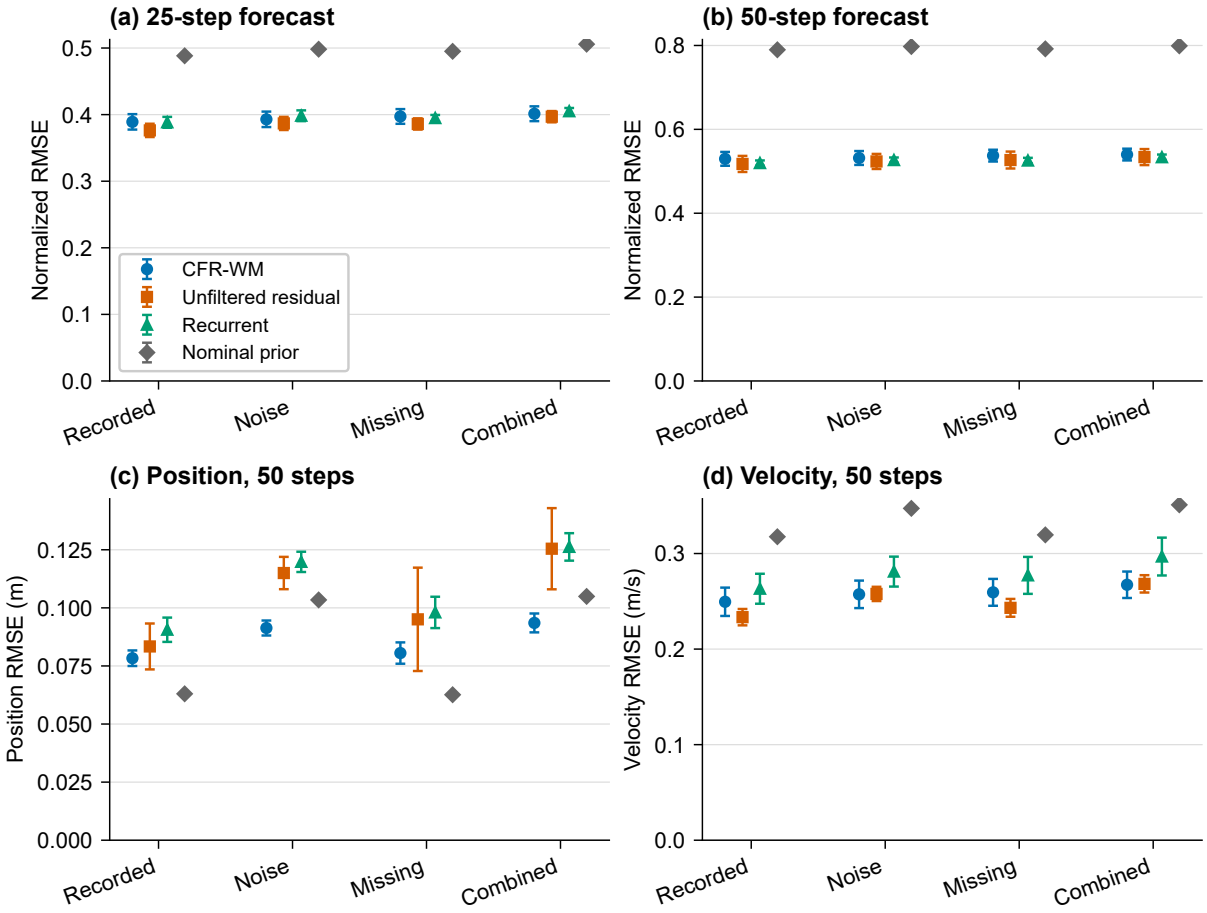


Fig. 4 Prediction on the held-out Melon flights: both normalized-error horizons and 50-step position and velocity errors. Error bars describe training-seed variation on the same three test flights.

Table A1 Controlled-system definitions. u is scalar in every system.

System	State	Δt	Nominal parameters
Damped oscillator	$(q, \dot{q})$	0.050	$m = 1.0,\ k = 1.6,\ c = 0.18$
Forced Duffing	$(q, \dot{q})$	0.040	$\alpha = -1.0,\ \beta = 1.0,\ c = 0.25$
Controlled Van der Pol	$(q, \dot{q})$	0.035	$\mu = 1.2,\ g_u = 1.0$
Torque pendulum	$(\sin\theta, \cos\theta, \dot{\theta})$	0.040	$g = 9.81,\ l = 1.0,\ m = 1.0,\ c = 0.12$
Cart–pole	$(x, \dot{x}, \theta, \dot{\theta})$	0.025	$g = 9.81,\ m_c = 1.0,\ m_p = 0.1,\ l/2 = 0.5,\ c = 0.04$

Appendix B Confirmatory grid

The prespecified grid comprises 11 methods, eight seeds (41, 53, 67, 79, 83, 97, 109, and 113), five systems, and three test regimes. It therefore contains 440 fitted models, 1,320 aggregate evaluation cells, and 126,720 trajectory-level rows. Each seed–system dataset contains 256 training, 64 validation, and 96 test trajectories of 72 time steps. Models receive eight context steps and predict 25 steps open loop. Training observations use noise equal to 3% of the fixed training-state scale and a nominal 10% independent missing probability; the corrupt regimes use 8% observation noise and a nominal 25% probability. The first time point of each generated trajectory is forced observed

Table A2 Complete independently sampled physical-parameter intervals. Symbols follow Table A1; g_u is the Van der Pol control gain.

System	Training interval	OOD interval
Damped oscillator	$m \in [0.90, 1.10],\ k \in [1.40, 1.80],\ c \in [0.14, 0.22]$	$m \in [1.25, 1.45],\ k \in [2.05, 2.35],\ c \in [0.26, 0.34]$
Forced Duffing	$\alpha \in [-1.10, -0.90],\ \beta \in [0.90, 1.10],\ c \in [0.20, 0.30]$	$\alpha \in [-0.75, -0.60],\ \beta \in [1.25, 1.45],\ c \in [0.34, 0.44]$
Controlled Van der Pol	$\mu \in [1.00, 1.40],\ g_u \in [0.90, 1.10]$	$\mu \in [1.65, 1.95],\ g_u \in [1.25, 1.45]$
Torque pendulum	$g \in [9.65, 9.95],\ l \in [0.90, 1.10],\ m \in [0.90, 1.10],\ c \in [0.08, 0.16]$	$g \in [10.25, 10.55],\ l \in [1.25, 1.45],\ m \in [1.25, 1.45],\ c \in [0.20, 0.28]$
Cart–pole	$g \in [9.65, 9.95],\ m_c \in [0.90, 1.10],\ m_p \in [0.085, 0.115],\ l/2 \in [0.45, 0.55],\ c \in [0.02, 0.06]$	$g \in [10.25, 10.55],\ m_c \in [1.25, 1.45],\ m_p \in [0.135, 0.160],\ l/2 \in [0.65, 0.78],\ c \in [0.075, 0.105]$

in every channel, with independent masks sampled at all subsequent trajectory times; a later sliding-window context can begin with an imputed value. ID-clean and ID-corrupt evaluations share trajectory identifiers; the OOD trajectory set is disjoint.

The five broad comparators are compact, protocol-matched RSSM-style, SimNorm-style, Transformer-style, PGNN-style, and PG-DMM-style transitions. Five prespecified CFR sensitivities test gains 0, 0.25, 0.75, and 1, plus a one-step-trained gain-0.5 model. CFR-WM retains the unused learned-gain-head parameters to maintain identical parameter counts across fixed-gain comparisons.

All methods ingest normalized context values, masks, log-one-plus elapsed times, and actions. Except for the Transformer-style encoder, context is encoded by a GRU; open-loop recurrent updates use a GRUCell. Every MLP has two hidden linear layers with SiLU activation. Table B3 lists the remaining architecture-specific choices.

For CFR-WM, observed context channels use gain 0.5 and missing channels use gain 0 exactly. The learned gain head remains instantiated but is inactive. The four gain sensitivities change only the observed-channel gain; the one-step sensitivity changes only the likelihood horizon.

Appendix C Hyperparameter search

The same six learning-rate/weight-decay trials were run on development seed 11 for every system–baseline pair (Table C4). The stored reliability-loss weight is shown because it is part of the generic training configuration, but it is inactive for all confirmatory methods in this paper. Each baseline uses the validation-NLL winner in Table C5; CFR-WM and all CFR sensitivities inherit the PGNN-style winner for the same system.

Appendix D Confirmatory estimands and decisions

H1 is the sole confirmation gate. It asks whether CFR-WM improves fixed-training-scale NRMSE-25 over the preselected PGNN-style comparator by at least 5%. Within each seed, PGNN-style and CFR-WM are averaged with equal weight over all 15 fixed system–regime cells; the primary point estimate is the arithmetic mean of the eight seed-specific relative improvements. H1 requires finite metrics, a 95% bootstrap lower endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the 5% margin.

H2 is a hierarchically gated robustness claim. Conditional on H1, the combined corrupt and parameter-shifted regimes must satisfy the same 5% improvement rule, while ID-clean performance must be noninferior under a signed relative margin of -5%. Both components require their corresponding bootstrap lower endpoint to meet or exceed the margin and a one-sided exact sign-flip $p < 0.05$. If H1 fails, H2 is not supported irrespective of its component estimates.

The bootstrap uses 5,000 replicates. Each replicate resamples the eight seed clusters globally with replacement and then resamples paired trajectory identifiers within every retained seed–system block. A single sampled identifier vector is reused for ID-clean and ID-corrupt repeated measurements, whereas the disjoint OOD identifiers are sampled separately. All five systems and three regimes remain in every replicate, and baseline, candidate, and their relative ratio are recomputed within the replicate. Exact tests enumerate all 2^8 seed-level sign flips and therefore rely on a seed-effect symmetry assumption. The five broad-comparator contrasts and the five sensitivity contrasts form separate Holm-adjusted families. The 15 system–regime contrasts are descriptive and carry no cell-level significance claim.

Table B3 Exact compact transition architectures used in the confirmatory grid. The bounded correction is $0.25\tanh(\cdot)$ and all forecast covariances are diagonal.

Method	Width	Context representation	Forecast transition and variance
RSSM-style	80	GRU; sigmoid feature scaling	data-only MLP mean/log-variance; GRUCell rollout
SimNorm-style	80	GRU; eight-group simplex projection	data-only MLP mean/log-variance; GRUCell rollout
Transformer-style	48	two causal layers, four heads, FF width 96, GELU, dropout 0	data-only MLP mean/log-variance; GRUCell rollout
PGNN-style	84	GRU	nominal one-substep RK4 plus deterministic residual MLP; learned constant log-variance
PG-DMM-style	84	GRU	nominal one-substep RK4 plus residual mean/log-variance MLP
CFR-WM and gain sensitivities	72	GRU states plus mask-gated fixed-gain context update	nominal one-substep RK4 plus residual mean/log-variance MLP

Table C4 Six-trial optimization grid.

Trial	Learning rate	Weight decay	Stored λ_{rel}
0	3×10^{-4}	10^{-6}	0.1
1	6×10^{-4}	10^{-5}	0.2
2	10^{-3}	10^{-5}	0.2
3	1.5×10^{-3}	10^{-4}	0.3
4	2×10^{-3}	10^{-5}	0.4
5	3×10^{-3}	10^{-4}	0.2

Appendix E Secondary analysis protocol

The classical baseline applies the same normalized nominal step and state projection as CFR-WM. Its unfiltered version starts from the final causally imputed context state; its filtered version applies gain 0.5 to each observed channel throughout the eight-step context. Both then use nominal dynamics only for all 25 forecast steps. The residual-scale analysis loads every trained CFR-WM checkpoint and changes only the multiplier on $\tanh(\mathbf{r})$ to 0, 0.125, 0.25, or 0.5. The stored 0.25 aggregate is reproduced with maximum absolute difference 0.00e+00.

The structured-outage audit starts from clean ID windows and makes all channels missing for context pair 3–4, 5–6, or 7–8. The previous observation is carried forward and elapsed time is set to one and two steps. Results average the three positions, five systems, and fixed cells within each of the eight seeds. The projection audit selects missing sine/cosine entries in the corrupt ID and corrupt OOD pendulum contexts and measures the absolute difference between the preprojection and projected normalized states.

Stored latency is computed separately for every method, seed, system, and regime. One batch contains all 96 evaluation trajectories. Timing begins before the batch tensors move from CPU to GPU, synchronizes CUDA after the 25-step rollout, and excludes data-loader iteration and metric computation. The value is divided by 96 before seed-balanced aggregation. It is a batched throughput measure.

External uncertainty is recomputed from the three held-out flight predictions for all five final seeds of CFR-WM, the unfiltered residual model, and the data- only recurrent model. Flight-level NLL, nominal 90% marginal coverage, mean interval width, and central-interval calibration error are first given equal flight weight and then summarized across seeds. Recomputed prediction means match the stored means with maximum absolute difference 0.00e+00.

Appendix F Reproducibility checks

The reproducibility contract records the method list, seeds, data design, training configuration, decision rules, and SHA-256 hashes of the training, evaluation, analysis, and tuning artifacts. Analysis requires the complete grid, finite metrics, trajectory pairing, disjoint OOD identifiers, 40 distinct seed– system data hashes, and matching stored provenance. The top-level independent units are the eight stochastic training/data realizations, with trajectories nested within them. Statistical inference concerns the prespecified finite benchmark and variation across those seed realizations.

Appendix G Complete descriptive cell contrasts

Table G6 reports every prespecified CFR-WM–PGNN-style system–regime contrast. Absolute intervals come from the paired nested bootstrap. Relative changes use PGNN-style as the denominator, with negative values favoring CFR-WM. These fixed-cell estimates are descriptive.

Table C5 Selected trial index by system and protocol-matched baseline.

System	RSSM-style	SimNorm-style	Transformer-style	PGNN-style	PG-DMM-style
Damped oscillator	3	2	4	5	1
Forced Duffing	5	4	5	5	5
Controlled Van der Pol	5	5	5	5	0
Torque pendulum	5	5	5	5	4
Cart–pole	5	5	5	5	5

Table G6 All 15 descriptive CFR-WM versus PGNN-style NRMSE-25 contrasts.

System	Regime	CFR–PGNN-style [95% CI]	Relative change, %
Cartpole	ID clean	+0.0163 [+0.0103, +0.0225]	+47.33
Cartpole	ID corrupt	-0.0207 [-0.0281, -0.0136]	-13.33
Cartpole	OOD + corrupt	-0.0111 [-0.0194, -0.0027]	-5.66
Controlled Van Der Pol	ID clean	+0.0059 [-0.0022, +0.0149]	+19.36
Controlled Van Der Pol	ID corrupt	-0.0318 [-0.0381, -0.0250]	-32.39
Controlled Van Der Pol	OOD + corrupt	-0.0083 [-0.0148, -0.0016]	-4.74
Damped Oscillator	ID clean	-0.0011 [-0.0041, +0.0019]	-2.41
Damped Oscillator	ID corrupt	-0.0291 [-0.0341, -0.0244]	-32.12
Damped Oscillator	OOD + corrupt	-0.0257 [-0.0295, -0.0223]	-27.86
Forced Duffing	ID clean	+0.0094 [+0.0028, +0.0168]	+24.37
Forced Duffing	ID corrupt	-0.0295 [-0.0342, -0.0249]	-30.24
Forced Duffing	OOD + corrupt	+0.0094 [+0.0011, +0.0174]	+3.90
Torque Pendulum	ID clean	-0.0046 [-0.0082, -0.0005]	-8.59
Torque Pendulum	ID corrupt	-0.0378 [-0.0436, -0.0316]	-34.80
Torque Pendulum	OOD + corrupt	+0.0110 [+0.0055, +0.0165]	+5.94

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